9.

Invisible Light

And therefore as a stranger give it welcome.
There are more things in heaven and earth, Horatio,
Than are dreamt of in your philosophy

HAMLET, ACT 1, SCENE 5

Before 1800 the word "light," apart from its use as a verb and an adjective, referred just to visible light. But early that year the English astronomer William Herschel observed some warming that could only have been caused by a form of light invisible to the human eye. Already an accomplished observer, Herschel had discovered the planet Uranus in 1781 and was now exploring the relation between sunlight, color, and heat. He began by placing a prism in the path of a sunbeam. Nothing new there. Sir Isaac Newton had done that back in the 1600s, leading him to name the familiar seven colors of the visible spectrum: red, orange, yellow, green, blue, indigo, and violet. (Yes, the colors do indeed spell Roy G. Biv.) But Herschel was inquisitive enough to wonder what the temperature of each color might be. So he placed thermometers in various regions of the rainbow and showed, as he suspected, that different colors registered different temperatures.[†]

Well-conducted experiments require a "control"—a measurement where you expect no effect at all, and which serves as a kind of idiot-check on what you are measuring. For example, if you wonder what effect beer has on a tulip plant, then also nurture a second tulip plant, identical to the first, but give it water instead. If both plants die—if you killed them both—then you can't blame the alcohol. That's the value of a control sample. Herschel knew this, and laid a thermometer outside of